

SQL Exploratory Data Analysis (EDA) Project

1. Introduction

In a competitive e-commerce environment, businesses rely on data to understand what drives revenue, how customers behave, and which products contribute the most to overall performance. This project presents an Exploratory Data Analysis (EDA) of an e-commerce sales dataset using SQL, with the goal of uncovering actionable insights that can support strategic, operational, and marketing decisions.

2. Business Background

The business operates in the e-commerce retail space, offering a diverse catalog of products across multiple categories, including Bikes, Components, Clothing, and Accessories. Sales are made to customers across several countries, with a strong presence in markets such as the United States, Australia, Canada, and the United Kingdom.

The company's data is structured using a star schema, consisting of:

- A Sales fact table capturing transactional details such as order dates, quantities, prices, and revenue
- Customer and Product dimension tables containing demographic and product classification information

Management requires a clear understanding of:

- Core revenue drivers
- Customer and product concentration risks
- Market performance across regions
- Opportunities for improving profitability and customer lifetime value

This EDA was conducted to provide stakeholders with data-driven visibility into these areas, enabling informed decision-making before investing in advanced analytics, forecasting, or optimization initiatives.

3. Tools Used

To complete this project, I used the following tools:

- SQL – Core tool for data exploration, joins, aggregations, and analysis
- PostgreSQL – Relational database used to store and query the data
- pgAdmin / VS Code – For writing and executing SQL queries

4. Data Model Overview

The database follows a star schema, commonly used in analytics systems.

Tables Used:

fact_sales

- Contains transactional sales data
- Metrics: quantity sold, revenue, discounts, order date

dim_customers

- Customer attributes such as segment, country, and region

dim_products

- Product details including category, price, and product name

Relationships

- fact_sales.customer_key → dim_customers.customer_key
- fact_sales.product_key → dim_products.product_key

This structure enables complex joins, aggregations, and performance analysis across multiple dimensions.

5. The Analysis

Each SQL query in this project was designed to answer a specific business question.

1. Dimensions Exploration: Customers & Products

Business Question

Before analyzing sales performance, the business needs to understand how its data is structured:

- Which countries do customers come from?
- How are products organized across categories and subcategories?
- What dimensions can be used later to segment revenue, customers, and performance?

This step helps stakeholders understand market coverage and product assortment before deeper analysis.

Approach

To answer these questions, I explored the dimension tables (dim_customers and dim_products) to identify:

- Low-cardinality dimensions (e.g. country, category) useful for grouping and dashboards

- High-cardinality dimensions (e.g. product names) useful for detailed drill-downs

I used SELECT DISTINCT queries to list unique values in each dimension. This is a standard EDA technique used to:

- Validate data consistency
- Identify segmentation opportunities
- Prepare for joins with the fact table (fact_sales)

SQL Queries Used

```
--Explore Customer Countries
SELECT
  DISTINCT country
FROM
  dim_customers
;
```

country
abc Filter...
United Kingdom
n/a
Australia
Germany
Canada
France
United States

```
--Explore Product Categories & Hierarchy
SELECT
  DISTINCT category,
    subcategory,
    product_name
FROM
  dim_products
ORDER BY
  category, subcategory
;
```

category	subcategory	product_name
abc Filter...	abc Filter...	abc Filter...
Accessories	Bike Racks	Hitch Rack - 4-Bike
Accessories	Bike Stands	All-Purpose Bike Stand
Accessories	Bottles and Cages	Mountain Bottle Cage
Accessories	Bottles and Cages	Road Bottle Cage
Accessories	Bottles and Cages	Water Bottle - 30 oz.
Accessories	Cleaners	Bike Wash - Dissolver
Accessories	Fenders	Fender Set - Mountain
Accessories	Helmets	Sport-100 Helmet- Black
Accessories	Helmets	Sport-100 Helmet- Blue
Accessories	Helmets	Sport-100 Helmet- Red
Accessories	Hydration Packs	Hydration Pack - 70 oz.
Accessories	Lights	Headlights - Dual-Beam
Accessories	Lights	Headlights - Weatherpro...
Accessories	Lights	Taillights - Battery-Power...

Key Outcomes & Insights

1. Customer Geography

Customers are distributed across multiple countries, including:

- United States
- United Kingdom
- Canada
- Germany
- France
- Australia

This confirms the business operates in a multi-country market, making geographic segmentation essential for later revenue and performance analysis.

Country is a low-cardinality, high-value dimension ideal for:

- Sales by country
- Regional dashboards
- Market comparison analysis

2. Product Structure & Assortment

Products are organized into clear hierarchical levels:

- Category → Subcategory → Product

Categories such as Components and Bikes contain significantly more products than others.

This structure enables:

- Category-level performance analysis
- Identification of over- or under-represented product lines
- Scalable product reporting in dashboards

Why This Matters for the Business

This dimension exploration ensures that:

- The dataset is well-structured and analysis-ready
- Stakeholders can confidently segment performance by:
 - Geography
 - Product category
- Future analyses (revenue, customer value, trends) are built on validated dimensions, reducing the risk of misleading insights

This step lays the foundation for all upcoming sales, customer, and product performance analysis.

2. Date & Time Exploration

Business Question

Before analyzing trends and performance, the business needs to understand:

- What is the time range of the sales data?
- Is the dataset large enough to support year-over-year and trend analysis?
- What is the age range of customers represented in the data?

Understanding these boundaries helps avoid misleading conclusions and ensures trends are interpreted correctly.

Approach

I explored date-related fields in both the fact and dimension tables by:

- Identifying the earliest and latest order dates to determine the sales timeline
- Calculating the total number of years covered by the dataset
- Analyzing customer birthdates to understand demographic coverage

This step ensures the data has sufficient historical depth for meaningful analysis.

SQL Queries Used

```
-- Find the date of the first and last order?  
-- How many years of sales are available?
```

```

SELECT
    MIN(order_date) AS first_order_date,
    MAX(order_date) AS last_order_date,
    DATE_PART('year', AGE(MAX(order_date), MIN(order_date))) AS
order_range_years
FROM
    fact_sales
;

```

first_order_date	last_order_date	order_range_years
abc Filter...	abc Filter...	abc Filter...
2010-12-29	2014-01-28	3

```

--Find the youngest and oldest customer
SELECT
    MIN(birthdate) AS oldest_birthdate,
    DATE_PART('year', AGE(NOW(), MIN(birthdate))) AS oldest_age,
    MAX(birthdate) AS youngest_birthdate,
    DATE_PART('year', AGE(NOW(), MAX(birthdate))) AS youngest_age
FROM
    dim_customers
;

```

oldest_birthdate	oldest_age	youngest_birthdate	youngest_age
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1916-02-10	109	1986-06-25	39

Key Outcomes

The sales data spans multiple years, making it suitable for:

- Trend analysis
- Seasonal comparisons
- Growth evaluation over time

The customer base includes a wide age range, indicating:

- Diverse customer demographics
- Potential for age-based segmentation in future analysis

Why This Matters

By confirming the time coverage and customer age distribution, stakeholders can trust that:

- Observed sales trends are not based on a short or incomplete dataset
- Demographic insights are representative and actionable

This validation step strengthens the reliability of all subsequent analyses.

3. Measures Exploration (Key Business Metrics)

Business Question

At the highest level, business stakeholders want to understand:

- How much revenue the business has generated?
- How many orders, customers, and products are involved?
- What the average selling price looks like?

These core KPIs provide a baseline view of business performance before deeper segmentation.

Approach

I calculated high-level aggregate metrics from the fact and dimension tables to establish:

- Overall sales scale
- Customer and product coverage
- Order volume and pricing context

This analysis intentionally avoids segmentation to focus on “big picture” performance.

SQL Queries Used

```
-- Find out the following multiple measures
```

```
--Find the Total Sales
```

```
SELECT SUM(sales_amount) AS total_sales FROM fact_sales;
```

```
--output: 29356250
```

```
--Find how many items are sold
```

```
SELECT SUM(quantity) AS total_quantity FROM fact_sales;
```

```
--output: 60423
```

```
--Find the average selling price
```

```
SELECT AVG(price) AS avg_price FROM fact_sales;
```

```
--output: 486.04
```

```
--Find the Total number of Orders
```

```
SELECT COUNT(order_number) AS total_orders FROM fact_sales;
```

```
--output: 60398
```

```
SELECT COUNT(DISTINCT order_number) AS total_orders FROM fact_sales;  
--output: 27659
```

```
--Find the Total number of Products  
SELECT COUNT(product_name) AS total_products FROM dim_products;  
--output: 295
```

```
SELECT COUNT(DISTINCT product_name) AS total_products FROM dim_products;  
--output: 295
```

```
--Find the Total number of Customers  
SELECT COUNT(customer_key) AS total_customers FROM dim_customers;  
--output: 18484
```

```
SELECT COUNT(DISTINCT customer_key) AS total_customers FROM dim_customers;  
--output: 18484
```

```
--Find the Total number of Customers that has placed an order  
SELECT COUNT(customer_key) AS total_customers FROM fact_sales;  
--output: 60398
```

```
SELECT COUNT(DISTINCT customer_key) AS total_customers FROM fact_sales;  
--output: 18484
```

Key Metrics Calculated (Results)

- Total Sales: \$29,356,250
- Total Items Sold: 60,423
- Average Selling Price: \$486.04
- Total Orders: 27,659
- Total Products: 295
- Total Customers: 18,484

A consolidated KPI query was also created using UNION ALL to produce a single metrics report, suitable for dashboards and executive summaries.

```
-- Generate Report that shows all key metrics of the business  
SELECT 'Total Sales' AS measure_name, SUM(sales_amount) AS measure_value  
FROM fact_sales  
UNION ALL  
SELECT 'Total Quantity', SUM(quantity) FROM fact_sales  
UNION ALL  
SELECT 'Average Price', ROUND(AVG(price), 2) FROM fact_sales  
UNION ALL  
SELECT 'Total No. Orders', COUNT(DISTINCT order_number) FROM fact_sales  
UNION ALL
```

```

SELECT 'Total No. Products', COUNT(DISTINCT product_name) FROM dim_products
UNION ALL
SELECT 'Total No. Customers', COUNT(DISTINCT customer_key) FROM
dim_customers
;

```

measure_name	measure_value
abc Filter...	abc Filter...
Total Sales	29356250
Total Quantity	60423
Average Price	486.04
Total No. Orders	27659
Total No. Products	295
Total No. Customers	18484

Key Outcomes & Insights

- The business has generated ~\$29.3M in total revenue, indicating a sizable transactional operation.
- With 27,659 orders and 18,484 customers, the data suggests a high proportion of one-time or low-frequency buyers, which is important for retention analysis.
- The average selling price (~\$486) provides useful context for:
 - Product positioning
 - Discount strategy evaluation
 - Revenue forecasting
- A product catalog of 295 items indicates enough variety to justify category-level and product-level performance analysis.

Why This Matters

These metrics serve as the foundation KPIs for all further analysis:

- They define the scale of the business
- They help validate data completeness
- They provide reference points for segmented insights (by country, product, customer, or time)

Without establishing these baseline measures, deeper analysis risks losing business context.

4. Magnitude Analysis (Category-Level Comparisons)

Business Question

After establishing overall KPIs, the business needs to understand where activity is concentrated:

- Which countries drive the customer base and sales volume?
- How is the customer base distributed by gender?
- Which product categories dominate inventory and revenue?
- Is revenue spread evenly across customers, or concentrated among a few?

These insights help prioritize markets, products, and customer segments.

Approach

I compared key measures across multiple dimensions by aggregating customers, products, revenue, and quantities sold.

This analysis focuses on relative importance, highlighting dominant segments rather than averages.

SQL Queries Used

```
-- Find total customers by countries
SELECT
    country,
    COUNT(customer_key) AS total_customers
FROM
    dim_customers
GROUP BY
    country
ORDER BY
    total_customers DESC
;
```

country	total_customers
abc Filter...	abc Filter...
United States	7482
Australia	3591
United Kingdom	1913
France	1810
Germany	1780
Canada	1571
n/a	337

```
-- Find the total customers by gender
SELECT
    gender,
    COUNT(customer_key) AS total_customers
FROM
    dim_customers
GROUP BY
    gender
ORDER BY
    total_customers DESC
;
```

gender	total_customers
abc Filter...	abc Filter...
Male	9341
Female	9128
n/a	15

```
-- Find total products by category
SELECT
    category,
    COUNT(product_key) AS total_products
FROM
    dim_products
GROUP BY
    category
ORDER BY
    total_products DESC
;
```

category	total_products
abc Filter...	abc Filter...
Components	127
Bikes	97
Clothing	35
Accessories	29
NULL	7

-- What are the average costs in each category?

```
SELECT
    category,
    ROUND(AVG(cost), 2) AS avg_cost
FROM
    dim_products
GROUP BY
    category
ORDER BY
    avg_cost DESC
;
```

category	avg_cost
abc Filter...	abc Filter...
Bikes	949.44
Components	264.72
NULL	28.57
Clothing	24.80
Accessories	13.17

-- What is the total revenue generated for each category?

```
SELECT
    p.category,
    SUM(f.sales_amount) AS total_revenue
FROM
    fact_sales AS f
LEFT JOIN dim_products AS p
    ON f.product_key = p.product_key
GROUP BY
    p.category
ORDER BY
    total_revenue DESC
;
```

category	total_revenue
abc Filter...	abc Filter...
Bikes	28316272
Accessories	700262
Clothing	339716

-- What is the total revenue generated by top 50 customer?

```
SELECT
  c.customer_key,
  c.first_name,
  c.last_name,
  SUM(f.sales_amount) AS total_revenue
FROM
  fact_sales AS f
LEFT JOIN dim_customers AS c
  ON f.customer_key = c.customer_key
GROUP BY
  c.customer_key,
  c.first_name,
  c.last_name
ORDER BY
  total_revenue DESC
LIMIT 50
;
```

customer_key	first_name	last_name	total_revenue
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1302	Nichole	Nara	13294
1133	Kaitlyn	Henderson	13294
1309	Margaret	He	13268
1132	Randall	Dominguez	13265
1301	Adriana	Gonzalez	13242
1322	Rosa	Hu	13215
1125	Brandi	Gill	13195
1308	Brad	She	13172
1297	Francisco	Sara	13164
434	Maurice	Shan	12914
440	Janet	Munoz	12488
242	Lisa	Cai	11468

-- What is the distribution of items sold across countries?

```
SELECT
  c.country,
  SUM(f.quantity) AS total_sold_items
FROM
  fact_sales AS f
LEFT JOIN dim_customers AS c
  ON f.customer_key = c.customer_key
GROUP BY
  c.country
ORDER BY
  total_sold_items DESC;
```

country	total_sold_items
abc Filter...	abc Filter...
United States	20481
Australia	13346
Canada	7630
United Kingdom	6910
Germany	5626
France	5559
n/a	871

Key Findings & Insights

1. Customer Distribution by Country:

- United States is the largest market with 7,482 customers, accounting for a significant share of the total customer base.
- Australia (3,591) and United Kingdom (1,913) follow, but at less than half and one-quarter of the U.S. size respectively.
- Remaining countries (France, Germany, Canada) contribute between 1,571–1,810 customers each.

Insight:

- Customer acquisition is heavily concentrated in the U.S., making it the primary market for growth, retention, and marketing optimization.

2. Customer Distribution by Gender

- Male customers: 9,341
- Female customers: 9,128
- Gender distribution is almost perfectly balanced, with no dominant group.

Insight:

- The business appeals broadly across genders, suggesting that performance differences are more likely driven by product or geography, not gender bias.

3. Product Distribution by Category

- Components: 127 products
- Bikes: 97 products
- Clothing: 35 products
- Accessories: 29 products

Insight:

- The catalog is heavily skewed toward Components and Bikes, indicating higher operational complexity and inventory depth in these categories.

4. Average Product Cost by Category

- Bikes: \$949.44 (highest average cost)
- Components: \$264.72
- Clothing: \$24.80
- Accessories: \$13.17

Insight:

- Bikes represent a high-cost, high-value category, requiring different pricing, discounting, and margin strategies compared to low-cost accessories and apparel.

5. Revenue Contribution by Category

- Bikes: \$28,316,272
- Accessories: \$700,262
- Clothing: \$339,716

Insight:

- Despite not having the largest product count, Bikes generate the overwhelming majority of total revenue, confirming them as the core revenue driver of the business.

This highlights a classic 80/20 dynamic between product categories.

6. Revenue Concentration: Top 50 Customers

- Top customers individually generate between ~\$8,500 and ~\$13,300 in revenue.
- The top 10 customers alone contribute ~\$130,000+ in combined revenue.

Insight:

- Revenue is highly concentrated among a small group of customers, emphasizing the importance of:
 - Retention strategies
 - Loyalty programs
 - Personalized engagement for high-value buyers

7. Items Sold by Country

- United States: 20,481 items sold
- Australia: 13,346

- Canada: 7,630
- United Kingdom: 6,910
- Germany: 5,626
- France: 5,559

Insight:

- Sales volume closely mirrors customer distribution, reinforcing the U.S. and Australia as the most operationally critical markets for inventory planning and logistics.

Why This Matters

Magnitude analysis reveals where the business truly makes money and where effort should be focused:

- The U.S. and Australia are priority markets
- Bikes are the primary revenue engine
- A small customer segment drives a disproportionate share of revenue

These insights directly inform marketing focus, inventory strategy, and customer retention initiatives.

5. Ranking Analysis (Top & Bottom Performers)

Business Question

Ranking analysis helps answer priority-driven questions, such as:

- Which products and customers generate the most revenue?
- Which items or customers contribute least to business performance?
- Where should the business double down, and where should it re-evaluate or optimize?

This analysis focuses on Top-N and Bottom-N performers across products and customers.

Approach

I ranked products and customers using revenue and order volume as key measures. Window functions were applied to assign ranks based on descending and ascending performance, allowing easy identification of extremes.

SQL Queries Used

```
-- Which 5 Products generate the highest revenue?
SELECT
  p.product_name,
  SUM(f.sales_amount) AS total_revenue,
  ROW_NUMBER() OVER(ORDER BY SUM(f.sales_amount) DESC) AS rank_products
FROM
  fact_sales AS f
LEFT JOIN dim_products AS p
  ON f.product_key = p.product_key
GROUP BY
  p.product_name
LIMIT 5
;
```

product_name	total_revenue	rank_products	
abc Filter...	abc Filter...	abc Filter...	
Mountain-200 Black- 46	1373454	1	
Mountain-200 Black- 42	1363128	2	
Mountain-200 Silver- 38	1339394	3	
Mountain-200 Silver- 46	1301029	4	
Mountain-200 Black- 38	1294854	5	

```
-- What are the 5 worst-performing products in terms of sales?
SELECT
  p.product_name,
  SUM(f.sales_amount) AS total_revenue,
  ROW_NUMBER() OVER(ORDER BY SUM(f.sales_amount) ASC) AS rank_products
FROM
  fact_sales AS f
LEFT JOIN dim_products AS p
  ON f.product_key = p.product_key
GROUP BY
  p.product_name
LIMIT 5
;
```

product_name	total_revenue	rank_products
abc Filter...	abc Filter...	abc Filter...
Racing Socks- L	2430	1
Racing Socks- M	2682	2
Patch Kit/8 Patches	6382	3
Bike Wash - Dissolver	7272	4
Touring Tire Tube	7440	5

```
-- Find the Top-10 customers who have generated the highest revenue?
SELECT
  c.customer_key,
  c.first_name,
  c.last_name,
  SUM(f.sales_amount) AS total_revenue,
  ROW_NUMBER() OVER(ORDER BY SUM(f.sales_amount) DESC) AS rank_customers
FROM
  fact_sales AS f
LEFT JOIN dim_customers AS c
  ON f.customer_key = c.customer_key
GROUP BY
  c.customer_key,
  c.first_name,
  c.last_name
LIMIT 10
;
```

customer_key	first_name	last_name	total_revenue	rank_customers
abc Filter...	abc Filter...	abc Filter...	abc Filter...	abc Filter...
1302	Nichole	Nara	13294	1
1133	Kaitlyn	Henderson	13294	2
1309	Margaret	He	13268	3
1132	Randall	Dominguez	13265	4
1301	Adriana	Gonzalez	13242	5
1322	Rosa	Hu	13215	6
1125	Brandi	Gill	13195	7
1308	Brad	She	13172	8
1297	Francisco	Sara	13164	9
434	Maurice	Shan	12914	10

```
-- Find the 3 customers with the fewest orders placed?
SELECT
  c.customer_key,
  c.first_name,
  c.last_name,
  COUNT(f.order_number) AS total_orders,
  ROW_NUMBER() OVER(ORDER BY SUM(f.sales_amount) ASC) AS rank_customers
FROM
  fact_sales AS f
LEFT JOIN dim_customers AS c
  ON f.customer_key = c.customer_key
GROUP BY
  c.customer_key,
  c.first_name,
  c.last_name
LIMIT 3
;
```

customer_key	first_name	last_name	total_orders	rank_customers
abc Filter...	abc Filter...	abc Filter...	abc Filter...	abc Filter...
17782	Alex	Collins	1	1
16994	Hunter	Miller	1	2
16993	Dylan	Taylor	1	3

Key Findings & Insights

1. Top 5 Revenue-Generating Products

Rank	Product Name	Total Revenue
1	Mountain-200 Black – 46	1,373,454
2	Mountain-200 Black – 42	1,363,128
3	Mountain-200 Silver – 38	1,339,394
4	Mountain-200 Silver – 46	1,301,029
5	Mountain-200 Black – 38	1,294,854

Insight:

All top 5 products belong to the Mountain-200 bike series, each generating ~\$1.3M+ in revenue individually.

This confirms:

- Strong product-market fit for high-end mountain bikes

- Revenue dominance driven by premium variants, not a wide product mix

Actionable takeaway:

Inventory planning, promotions, and supplier negotiations should heavily prioritize the Mountain-200 line.

2. Bottom 5 Products by Revenue

Rank	Product Name	Total Revenue
1	Racing Socks – L	2,430
2	Racing Socks – M	2,682
3	Patch Kit / 8 Patches	6,382
4	Bike Wash – Dissolver	7,272
5	Touring Tire Tube	7,440

Insight:

Low-priced accessory items generate less than 0.3% of the revenue of top bike products.

This suggests:

- These products are add-ons, not revenue drivers
- Possible issues with visibility, bundling, or demand

Actionable takeaway:

Instead of discontinuation, these items are better positioned as:

- Bundled products
- Cross-sell items during high-ticket purchases

3. Top 10 Customers by Revenue

Rank	Customer	Total Revenue
1	Nichole Nara	13,294
2	Kaitlyn Henderson	13,294
3	Margaret He	13,268
4	Randall Dominguez	13,265

5	Adriana Gonzalez	13,242
6	Rosa Hu	13,215
7	Brandi Gill	13,195
8	Brad She	13,172
9	Francisco Sara	13,164
10	Maurice Shan	12,914

Insight:

Top customers generate ~\$13K each, showing consistent high-value purchasing behavior rather than extreme outliers.

This reflects:

- Stable pricing
- Repeat purchasing of premium products
- No single customer dependency risk

Actionable takeaway:

These customers are ideal candidates for:

- Loyalty programs
- Early-access product launches
- Personalized retention strategies

4. Customers with the Fewest Orders

Rank	Customer	Total Orders
1	Alex Collins	1
2	Hunter Miller	1
3	Dylan Taylor	1

Insight:

These customers placed only one order each, representing low engagement or one-time purchases.

Actionable takeaway:

This segment is suitable for:

- Re-engagement campaigns
- First-time buyer incentives
- Onboarding or post-purchase follow-ups

Why This Ranking Analysis Matters

This analysis clearly shows that:

- Revenue is driven by a small set of premium bike products
- Accessories play a supporting role, not a primary revenue role
- Customer revenue is well-distributed among top buyers, reducing dependency risk
- There is a clear opportunity to convert one-time buyers into repeat customers

5. Conclusion:

1) Key Insights

This Exploratory Data Analysis revealed several clear and actionable patterns across sales, products, customers, and geography.

Revenue Concentration

- Total sales amount to \$29.36M, driven overwhelmingly by the Bikes category, which alone generated \$28.3M+ in revenue.
- Within Bikes, the Mountain-200 series dominates performance, with each top variant contributing ~\$1.3M individually.
- Accessories and Clothing contribute marginally and function primarily as supporting or add-on products rather than core revenue drivers.

Product Portfolio Structure

- The product catalog is largest in Components (127 products), yet this category does not translate into proportional revenue.
- High average product cost in Bikes (~\$949) aligns strongly with revenue dominance, confirming a premium-product business model.
- Low-revenue products such as socks, patch kits, and maintenance items show limited standalone impact and are better suited for bundling strategies.

Customer Behavior

- The customer base is evenly distributed by gender, indicating no gender-driven bias in purchasing behavior.
- Revenue among top customers is well distributed, with top buyers generating around \$13K each, reducing reliance on any single customer.
- A noticeable segment of customers placed only one order, highlighting opportunities for retention and repeat-purchase strategies.

Geographical Performance

- The United States is the largest market across customers, sales volume, and items sold.
- Australia, Canada, and the UK follow as strong secondary markets, showing potential for regional expansion and localized marketing.
- Sales volume closely mirrors customer distribution, indicating healthy market alignment rather than operational imbalance.

2) Closing Thoughts

This EDA successfully transformed raw transactional data into clear business insights by progressing from high-level metrics to detailed ranking and magnitude analysis.

The analysis demonstrates that:

- The business is premium-product driven, not volume-driven
- Revenue depends more on product quality and pricing power than catalog size
- Growth opportunities lie in customer retention, cross-selling, and regional optimization, not aggressive product expansion

From a technical standpoint, the project showcases:

- Strong use of aggregations, joins, window functions, and ranking logic
- A structured analytical flow from measures → magnitude → ranking
- The ability to connect SQL outputs to real business decisions

Overall, this EDA provides a solid analytical foundation that can support future work such as predictive modeling, customer segmentation, demand forecasting, or dashboard development.